

# Nuhiat Arefin

Bangladesh | [nuhiatarefin@gmail.com](mailto:nuhiatarefin@gmail.com) | [github.com/Nuhiat-Arefin](https://github.com/Nuhiat-Arefin)

I am an undergraduate junior studying Computer Science and Engineering at IUT. I enjoy using fuzzing to find real bugs, and I contribute fixes and tests to open-source projects.

## EDUCATION

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### Islamic University of Technology (IUT) | Bangladesh

*B.Sc. in Computer Science and Engineering | Partial scholarship*

### Notre Dame College | Dhaka, Bangladesh

*Higher Secondary Certificate (HSC)*

## OPEN-SOURCE EXPERIENCE

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### GSoC 2026 contributor, CCEXtractor

*Create an XDG Portal Backend for Regolith*

- Built and merged Rust features for wlroots/Wayland desktops, including background-app tracking and Remote Desktop clipboard sharing (PRs #199 and #214).
- Found a double-unref bug where xdg-desktop-portal could free the same object twice, corrupt memory, and crash. A maintainer acknowledged it, and I submitted PR #2111 with a regression test.
- Used zbus/D-Bus, systemd, Wayland, calloop/Tokio, and iced. Also tested two-way clipboard sharing between macOS and Linux with Deskflow.

[GSoC project](#) | [PR #199](#) | [PR #214](#) | [xdg-desktop-portal PR #2111](#)

## SELECTED RESEARCH & FUZZING

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### XVada: Toward Inferring Accurate Context-free Grammars for Big Languages in a Black-box Setting

arXiv preprint, 2026 | [arxiv.org/abs/2607.08959](https://arxiv.org/abs/2607.08959)

*Mohammad Rifat Arefin, Nuhiat Arefin, Shanto Rahman, Christoph Csallner*

- Co-authored a paper on learning grammars from program behavior, then used the generated Python Liquid grammar for fuzzing.
- Found and reported [CVE-2026-55865](#), an infinite loop that can hang the program, plus five other Python Liquid bugs that maintainers fixed.

### Miniscript Differential Fuzzing and Analysis | 2026

- Reported 33 issues across seven Miniscript implementations using differential fuzzing and focused analysis.
- Used differential fuzzing on NBitcoin, embit, and tinyminiscript to find cases where invalid scripts (including consensus-invalid ones) were accepted, deeply nested inputs crashed the program, or checks slowed down exponentially.
- Separately found rust-miniscript compiler crashes, a Bitcoin Core duplicate-key check that missed some branches, and several descriptor/parser issues.
- Reported an NBitcoin Derive() stack overflow that a project maintainer fixed, and authored two more NBitcoin fixes that were merged. Two rust-miniscript reports were fixed by other contributors; also contributed an ANTLR4 Miniscript grammar.

[NBitcoin stack fix](#) | [rust-miniscript #997](#) | [embit #144](#) | [tinyminiscript #62](#) | [Bitcoin Core #35629](#) | [ANTLR4 grammar](#)

**Merged fixes:** [NBitcoin #1323](#) | [NBitcoin #1324](#) | [rust-miniscript #1013](#) | [rust-miniscript #1016](#)

### llama.cpp / GGUF Fuzzing | 2026

- Used fuzzing to find an integer overflow while llama.cpp calculated GGUF tensor sizes. The bug could read past the end of memory; submitted draft PR #27200 with a fix.

[Upstream draft PR #27200](#)

## TECHNICAL SKILLS

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**Programming:** Rust, Go, C/C++, Python, Bash

**Fuzzing & Testing:** Grammar-based fuzzing, differential testing, AFL++, LibAFL, Nautilus, sanitizers, crash minimization, regression tests

**Systems & Tools:** Linux, container runtimes, OCI, Wayland, D-Bus, systemd, Git, CI